Spacecraft Attitude Determination and Controls System

CEC 300: Computing in Aviation and Aerospace

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Outline

- Overview
- Attitude Disturbances
- Stability control:
  - Passive Stabilization
  - Active Stabilization
- Control system (block diagram)

Introduction to Attitude Determination and Guidance System (ADGS)

- There are a variety of approaches to ADGS for spacecraft.
  - **<u>Passive methods:</u>**
    - Spin Stabilization
    - Gravity Gradient
    - Passive Magnetic
  - **<u>Active:</u>**
    - **Three-axis stabilization (our focus)**
      - A combination of actuators to control axes rotation and sensors to guide the spacecraft to determine its attitude and ensure a fixed orientation.
      - Active actuation previously covered in Module 2.
- **Challenge:** Spacecraft cannot leverage aerodynamic forces unlike aircraft.

Spacecraft Attitude Determination and Control System

Sensors

- **Sun Sensors**
  - Determines the Sun relative position
- **Star Trackers**
  - Estimate position based on known landmarks (Stars) within the imager's field of view
- **Inertial Measurement Units or Gyroscopes**
  - Indicates the angular rates about each axes.
- **Earth Horizon Sensor** – for missions where the satellite must point toward Earth, indicates the Earth's horizon relative to the spacecraft.

Like aircraft, space vehicles such as probes and satellites can be equipped with some combination of sensors and actuators sufficient to meet the mission requirements.

Actuators

- **Reaction Wheels**
  - controls rotational rate about an axes
- **Control Moment Gyroscopes** – enables rapid maneuvers of large spacecraft
- **Magnetorquers**
  - Provides torque about an axes commonly used for small satellite ADCS and desaturation of reaction wheels.
- **Reaction Control Thrusters**
  - Small thruster which uses gas expellant for attitude control about each axis.
  - Applications include detumbling, reaction wheel desaturation, and large maneuvers.

Pointing Accuracy

- Some payloads and other system components depend upon the aircraft pointing toward a target.

- Examples of pointing:
  - Communication antenna pointing
  - Image pointing toward a ground target, e.g. satellite imagery.
  - Images of celestial targets

- Pointing accuracy can be determined based on the diameter of the target and the distance to the target.
  - As h increases, ψ decreases.
  - As D decreases, ψ increases.
  - And, vice versa.

$$D = h\,\psi \qquad (4.3.1\text{-}1)$$

where

D = approximate diameter of target (m)

h = distance to target (m)

ψ = pointing accuracy (rad)

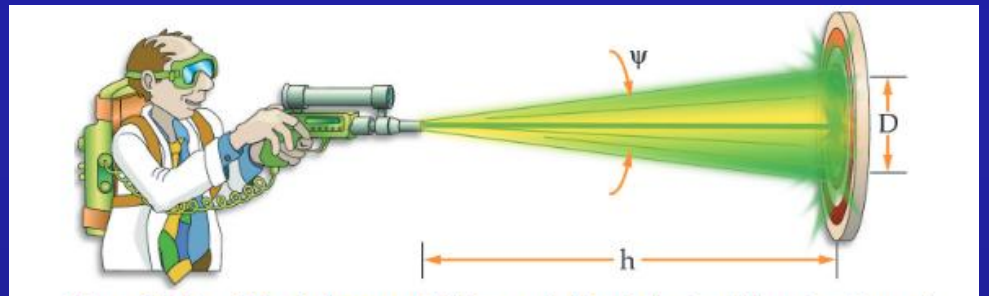

Figure 4.3.1-9. Attitude Accuracy. In this example, the shooter is pointing a laser beam at a dinner-plate-sized target. As his hand wavers, the beam describes a cone, more or less centered on the target. The angular size of this cone, ψ, defines attitude or pointing accuracy.

Table 4.3.1-1. Pointing Accuracy.

To point at a target the size of a dinner plate (25 cm or 10 in. diameter) at this distance...	The pointing accuracy needs to be...
1.4 m (4.6 ft.)	10°
14 m (46 ft.)	1°
140 m (460 ft.)	0.1°
1400 m (0.87 mi.)	0.01°

Attitude Dynamics

- $\vec{H}$, the vehicle's angular momentum is based upon the mass moment of inertia and the angular velocity. (see equation to right).
  - I is a measure of the object's resistance to changes in its rotational rate.

$$\vec{H} = I\vec{\Omega} \qquad (4.3.1\text{-}2)$$

where

$\vec{H}$ = angular momentum ($kg \cdot m^2/s$)

I = mass moment of inertia ($kg \cdot m^2$)

$\vec{\Omega}$ = angular velocity (rad/s)

- $\vec{\tau}$, or torque is applied to alter the acceleration about an axis.
  - "The magnitude of an applied torque depends on the force and distance
  - over which it is applied. The direction of torque is found using the
  - right-hand rule." - NASA, 2026.

$$\vec{T} = I\vec{\alpha} \qquad (4.3.1\text{-}3)$$

where

I = mass moment of inertia ($kg \cdot m^2$)

$\vec{\alpha}$ = angular acceleration ($rads/s^2$)

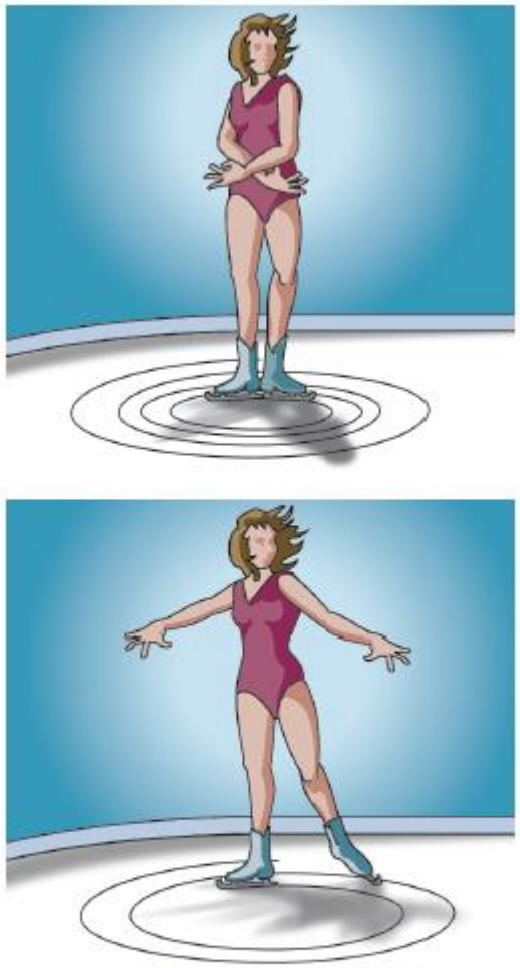

Figure 4.3.1-10. Changing Mass Moment of Inertia. Figure skaters change their moments of inertia to vary their rate of spin. For the same total angular momentum, they will spin faster by bringing their arms in (lower moment of inertia) and slower by extending their arms (higher moment of inertia).

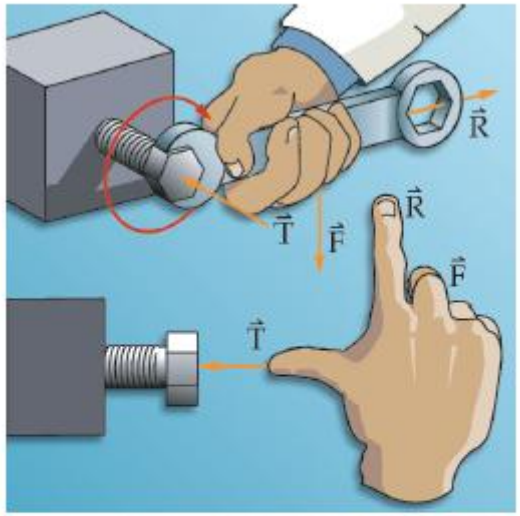

Figure 4.3.1-11. Torque. Turning a bolt with a wrench is a good example of applying a torque. You find the direction of torque using the right-hand rule. By wrapping the fingers of your right hand in the direction of spin, your thumb points in the direction of the torque vector. In this case, the torque direction is into the bolt, as we'd expect.

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Block Diagram of Dynamics for a Spacecraft

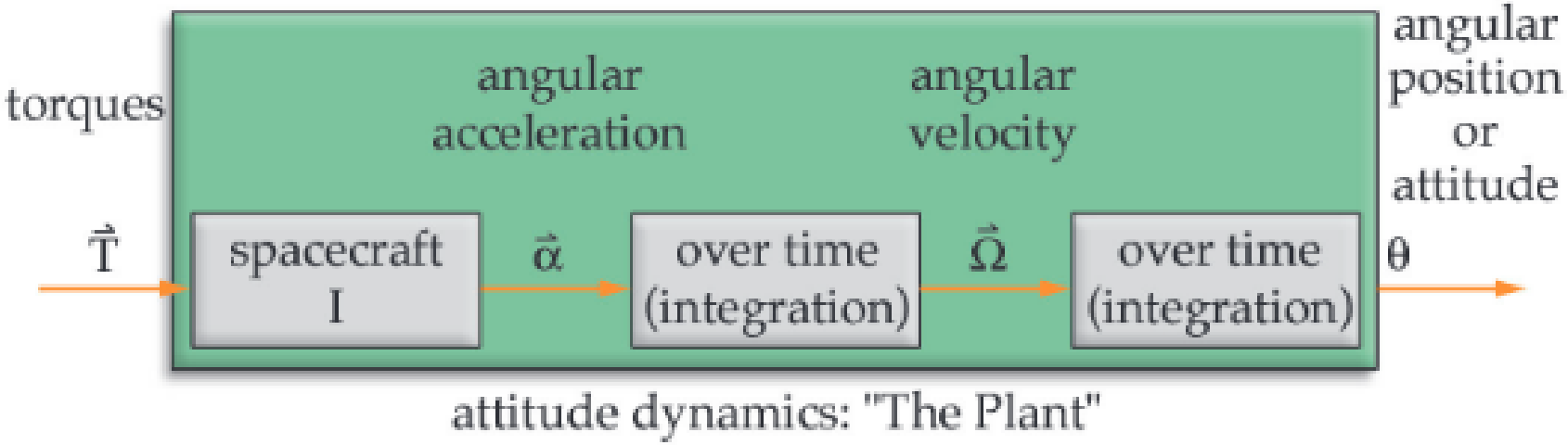

Figure 4.3.1-12. Block Diagram of Attitude Dynamics. A torque applied to a spacecraft (or any object for that matter) causes an angular acceleration. Over time, this acceleration increases angular velocity, changing its attitude.

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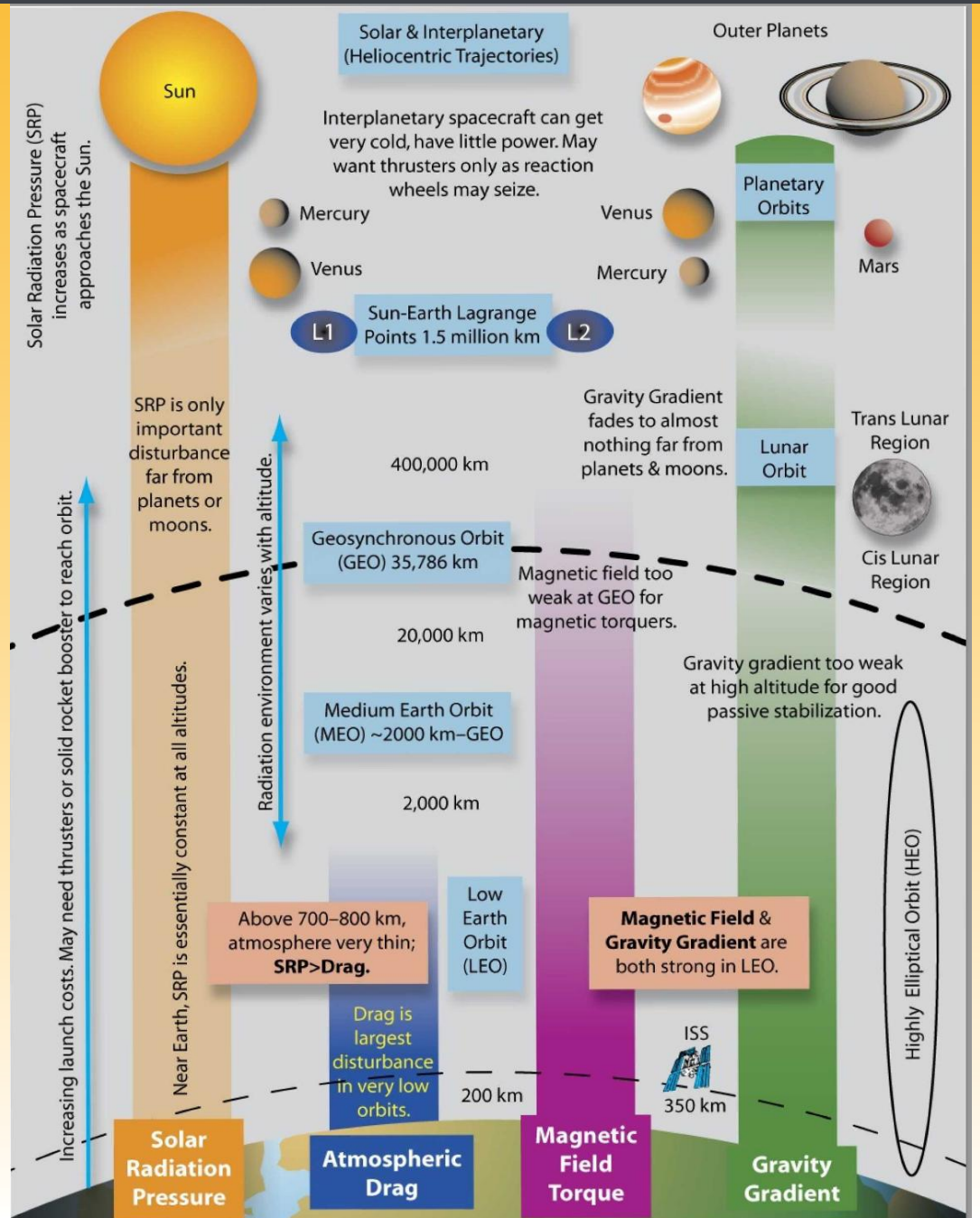
Altitude vs. Passive Forces / Torques for Vehicle Control



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Gravity Gradient Torque

$$\vec{F}_g = \frac{-\mu\, m}{R^2}\hat{R} \qquad (4.3.1\text{-}4)$$

where

$\vec{F}_g$ = force of gravity (N)

μ = gravitational parameter of the central body (km^3/s^2)

m = mass of the object (kg)

R = distance from the object to the central body (km)

$\hat{R}$ = unit vector in the $\vec{R}$ direction (dimensionless)

- The equation above shows the relationship of gravitational force, decreases at the rate of $\frac{1}{R^2}$
  - i.e. increasing R by twice its distance would result in $\frac{1}{4}$ of the gravitational force compared to its original length.
- When considering a dumbbell system with two vertically connected masses, when pointed Nadir, the lower mass experiences higher gravitational mass than the higher.
- In the case of a misalignment, the spacecraft is rotated back to vertical.

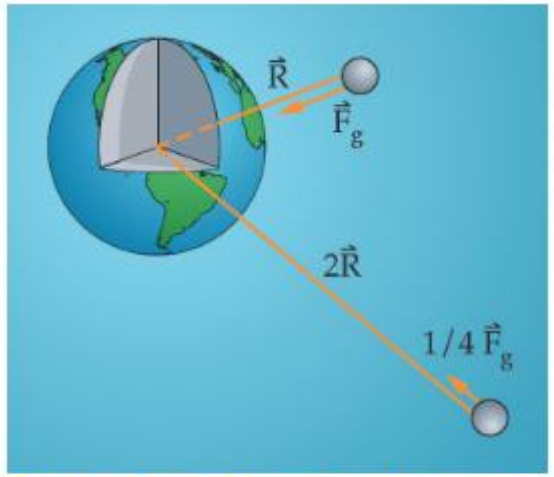


Figure 4.3.1-15. Gravitational Force. From Newton's Law of Universal Gravitation, we know that gravitational attraction decreases with the square of the distance between two objects. Thus, if we double the distance, the gravitational force is only 1/4 as strong.

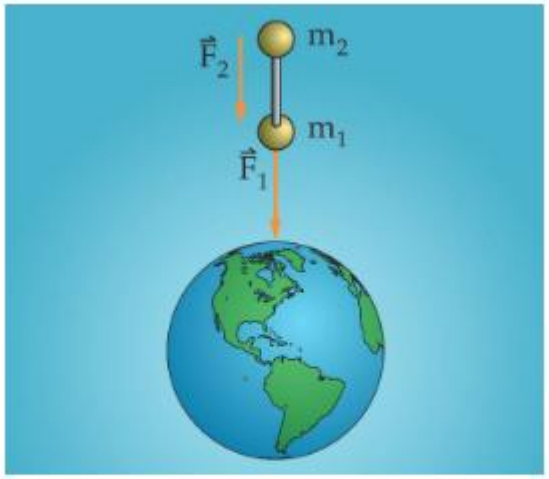


Figure 4.3.1-16. Gravity Gradient. In this simplified, dumbbell-shaped spacecraft, we show that the gravitational force on the lower part is slightly greater than the force on the upper part, $\vec{F}_1 > \vec{F}_2$. The same effect happens in more conventionally shaped spacecraft due to differences in internal mass distribution.

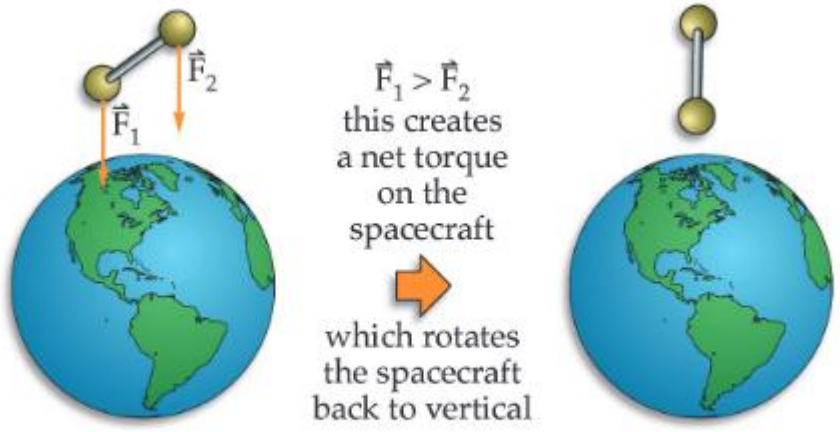


Figure 4.3.1-17. Gravity-gradient Torque. The slight difference in gravitational force between the upper and lower part of a spacecraft will tend to rotate the spacecraft to vertical, with its long axis pointed to Earth.

Solar-radiation Pressure

- Despite being massless, photons have momentum.
- When striking the spacecraft's surface, momentum is transferred.
  - Small force is granted only $9 * 10^{-15}$ N.
- The solar illuminated surface of the aircraft influences the amount of force against it. So, large solar panels will receive more force than a magnetometer boom.
- The moment arm is calculated by $T = F\ x\ d$ where F is the force on the surface and d is the distance from the center of mass.
  - If d=1m, then $T = 9 * 10^{-15}$ Nm.
- While small, the accumulation of these disturbances over time will result in a gradual increase in pointing error.

$$F = \left(\frac{Fs}{c}\right) A_s (1 + r) \cos I \qquad (4.3.1\text{-}6)$$

where

F = force on a surface (N)

Fs = solar constant = 1358 W/m^2 at Earth's orbit around the Sun

c = speed of light = 3×10^8 m/s

A_s = illuminated surface area (m^2)

r = surface reflectance (where r = 1 for a perfect reflector and 0 for a perfect absorber) (unitless)

I = incidence angle to the Sun (deg)

Magnetic Torque

- Based on the **effective dipole** of the spacecraft and the **strength of the local magnetic field**, the magnetic torque disturbances can be calculated using the equation to the right.
  - **Dipole** generated by the spacecraft derived from the accumulated charge produced from the vehicle impacting charged particles.
- Small satellites in LEO are more impacted than a large satellite in GEO.

$$T = D\,B \qquad (4.3.1\text{-}7)$$

where

T = torque on a spacecraft (Nm)

D = spacecraft dipole (amp-m^2)

B = local magnetic field's strength (tesla), which varies with altitude (R = distance to Earth's center) and latitude. Earth's magnetic moment, M, is about 7.99×10^{15} tesla-m^3. At the poles, $B = 2M/R^3$. At the equator, $B = M/R^3$. (tesla = kg / amp-s^2)

Aerodynamic Drag

- Drag from the aircraft's interactions with the upper atmosphere occurs in LEO.
  - Like solar-pressure, the **surface area of the impacted area**, A, correlates with the F_{drag}.
  - **Atmospheric density** ρ varies by altitude such that it decreases as altitude increases.
  - The **Coefficient of Drag,** C_D, must be determined:
    - A value of 2.0 to 2.5 is common for spacecraft.
    - Determined through
      - Satellite tracking and calibration
      - Numeric modelling / simulation
      - Empirical monitoring conducting experiments in an analogue to LEO conditions.
  - The **velocity**, v, significantly influences the drag results as its value is squared.

$$F_{drag} = \frac{1}{2} \rho V^2 C_D A \qquad (4.3.1\text{-}8)$$

where

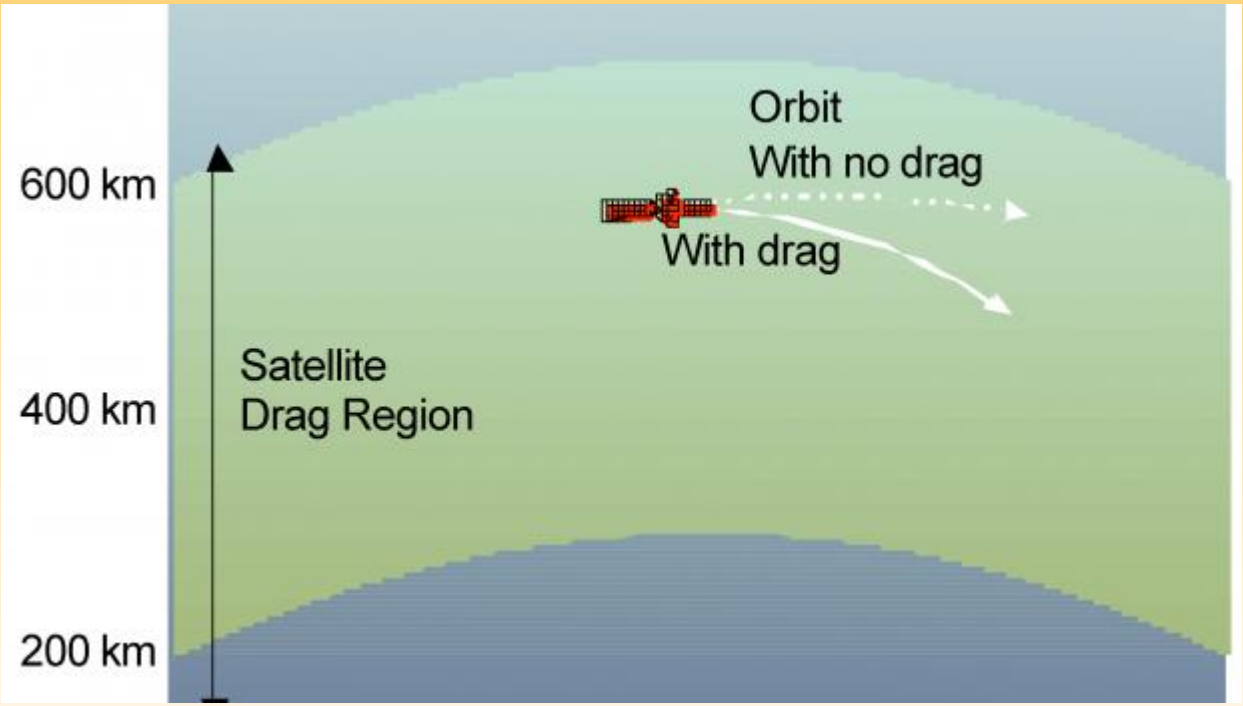
F_{drag} = force of drag (N)

ρ = atmospheric density (kg/m^3)

V = velocity (m/s)

C_D = drag coefficient (unitless)

A = impacted area (m^2)



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- Attitude Disturbances
- Stability control:
  - Passive Stabilization
  - Active Stabilization
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Stabilization approaches

- Passive actuation methods for stabilization are built into the satellite's design and operation.
  - Three will be discussed:
    - Spin Stabilization
    - Gravity-Gradient Stabilization
    - Dampers

- Active methods rely upon controlled actuators to influence attitude across a single or multiple axes.
  - The suite of sensors and actuators used for ADCS were listed earlier in this lesson and were discussed in Module 2 of the course.
  - These include:
    - Thrusters
    - Magnetic torquers
    - Momentum-control devices (includes Reaction Wheels)

Passive: Spin Stabilization

- This technique leverages the spin of the spacecraft at a constant angular-momentum.
- Result is a spacecraft that holds its inertial attitude.
- Example: Earth's H-vector points out of its north poll because of a constant angular rotation.
  - Stars seem to rotate about the Polaris (North Star) because our H axis points toward the Noth Star.
- Downside: Not good for pointing at the Earth

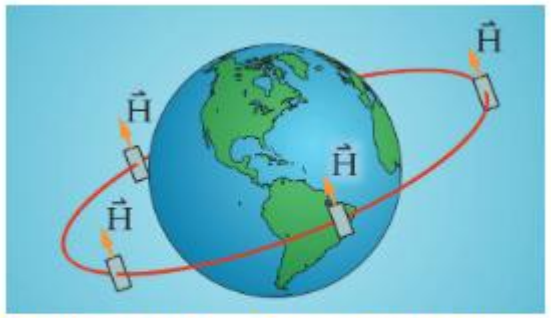


Figure 4.3.1-24. Spin Stabilization. A spinning spacecraft keeps its angular-momentum vector fixed in inertial space.

Figure 4.3.1-25. Spin Stabilization Isn't Much Good for Earth Pointing. Because spin stabilized spacecraft have fixed pointing with respect to inertial space, they aren't a good choice for Earth-pointing missions. During part of the orbit, they may point toward Earth but during other parts of the orbit, they'll point away.

- Dual-spin aircraft rely upon an inner "despun section" and an outer drum-like section that rotates at high angular rates.

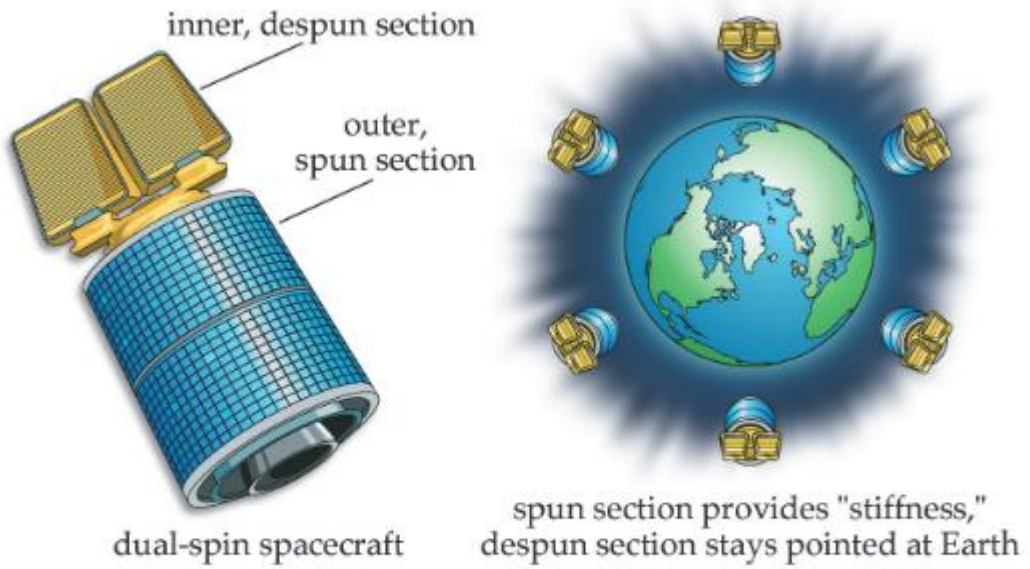
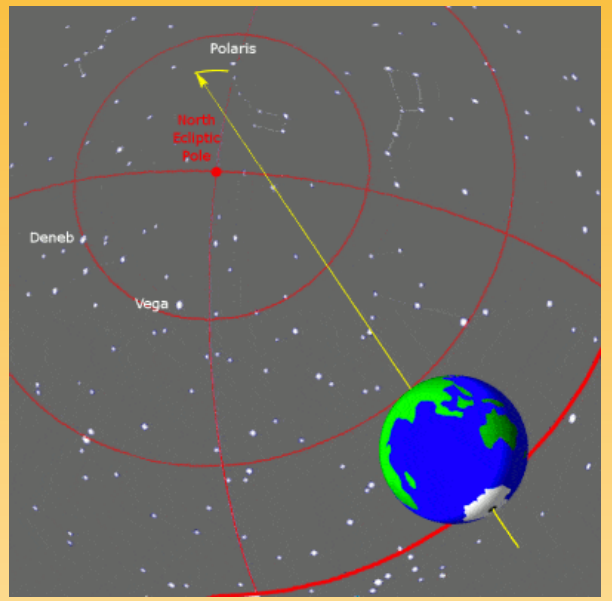


Figure 4.3.1-26. Dual-spin Spacecraft. A dual-spin spacecraft uses the inherent stiffness of a spinning outer section with a "de-spun" inner section that can independently point at Earth. The de-spun section turns at the orbital rate to keep sensors and antenna pointed at Earth.

Spin Stabilization Examples



- Star field appears to circle Polaris
- Through the spin stabilization of the Earth it seems to continually have its north axis aimed toward Polaris.
- Like a top, the pointing will move around in a circle, called precession.
  - While the axis currently points to Polaris, over time the pointing will shift to another celestial body
  - Period of about 26,000 years to move from the north star to other bodies and back.

The North Star: What's So Special About It? | AstroCamp

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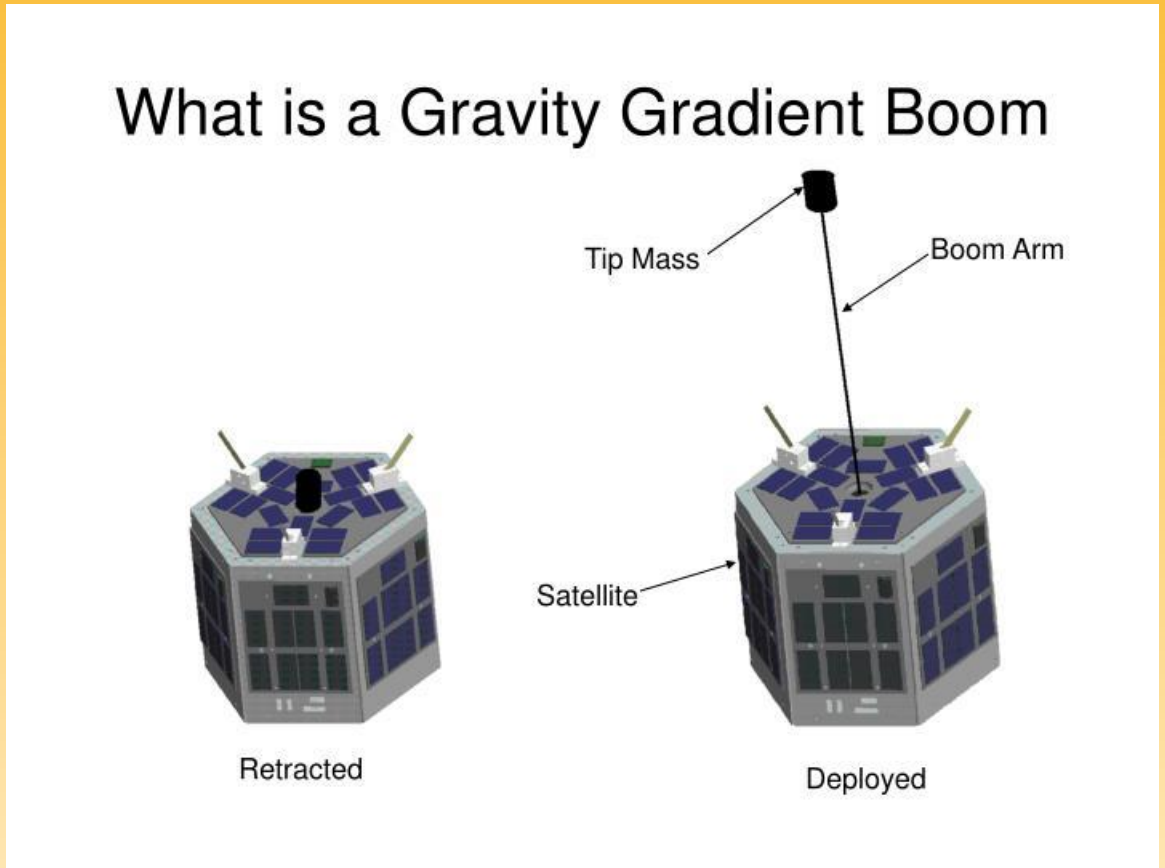
Passive: Gravity Gradient

- Exploiting the gravity gradient discussed earlier, this approach enables the spacecraft to point to the local downward.

- Just like the dumbbell visualization earlier, a mass at the end of the boom.

- Limitations:
  - Only pitch and yaw can be controlled
  - Limited accuracy, $\pm 10°$ (depends upon the moments of inertia)
  - Limited altitude – relies upon the forces of gravity, such that at higher altitudes this approach becomes less effective.

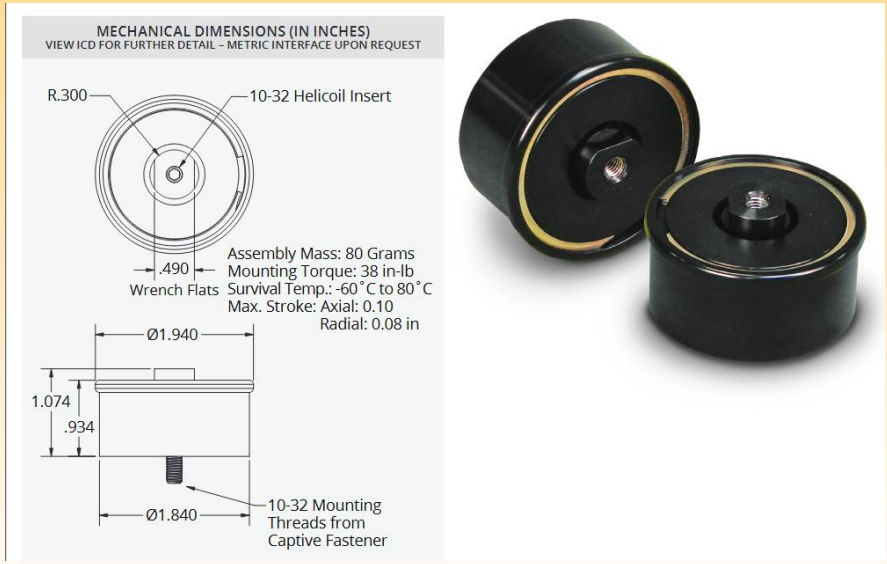


Dampers

- Decreases or dampens angular momentum through the absorption of energy.

- One approach discussed in NASA (accessed 2026) uses a ball inside of a circular tube filled with a highly viscous fluid.
  - Energy is dissipated resulting in a reduction in angular rate because of friction between the ball and the tube's fluid.

- Limits the rotational rate about a spin axis, which can be used to remove variability or wobble of angular motion about the spin axis.



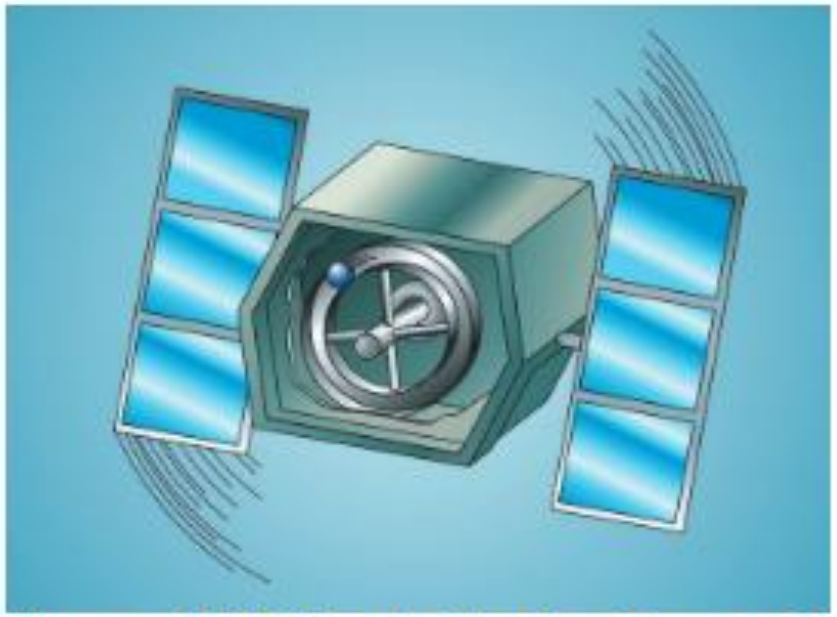


Figure 4.3.1-28. A Simple Spacecraft Damper. Dampers "absorb" unwanted angular momentum by converting the energy into friction, in much the same way as the brakes in a car turn linear momentum into heat through friction. A ball inside a circular tube filled with a viscous fluid is one type of damper. As the spacecraft rotates, the ball moves through the fluid. The resistance produces heat, dissipating the angular motion.

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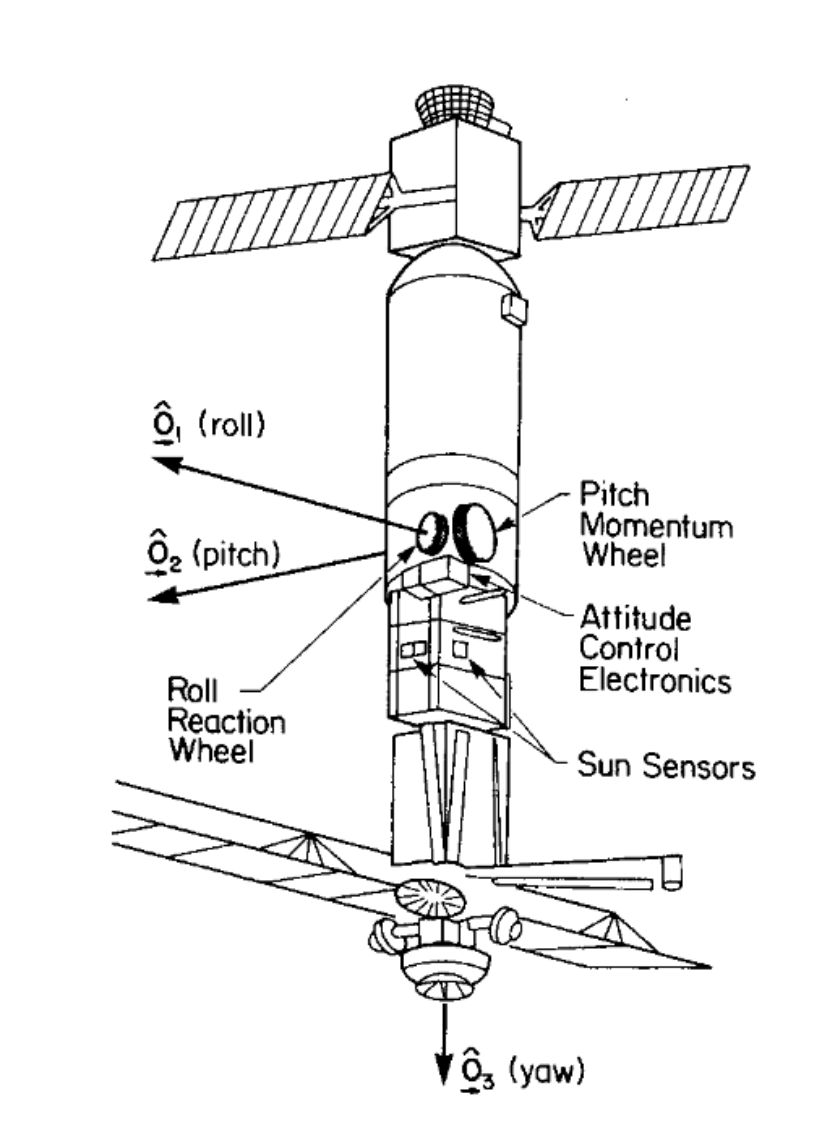

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Control System for ADCS

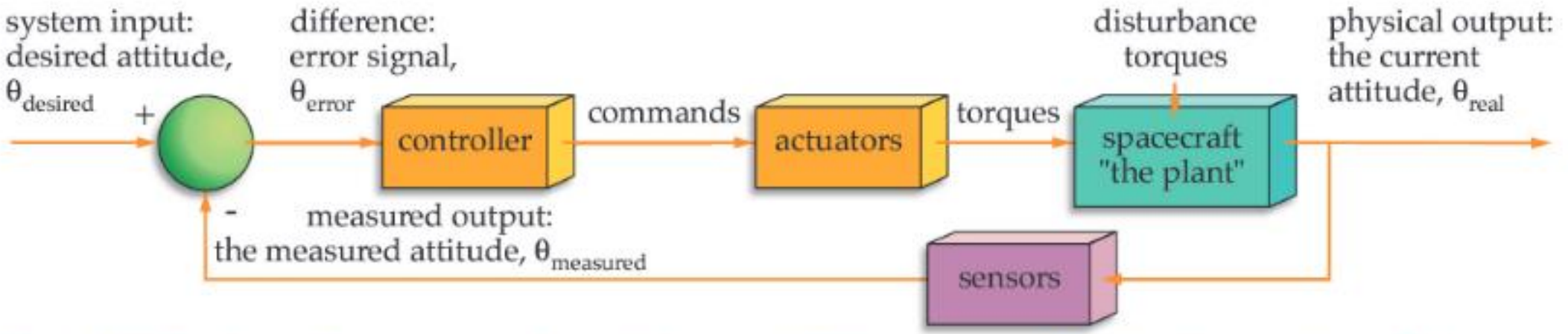
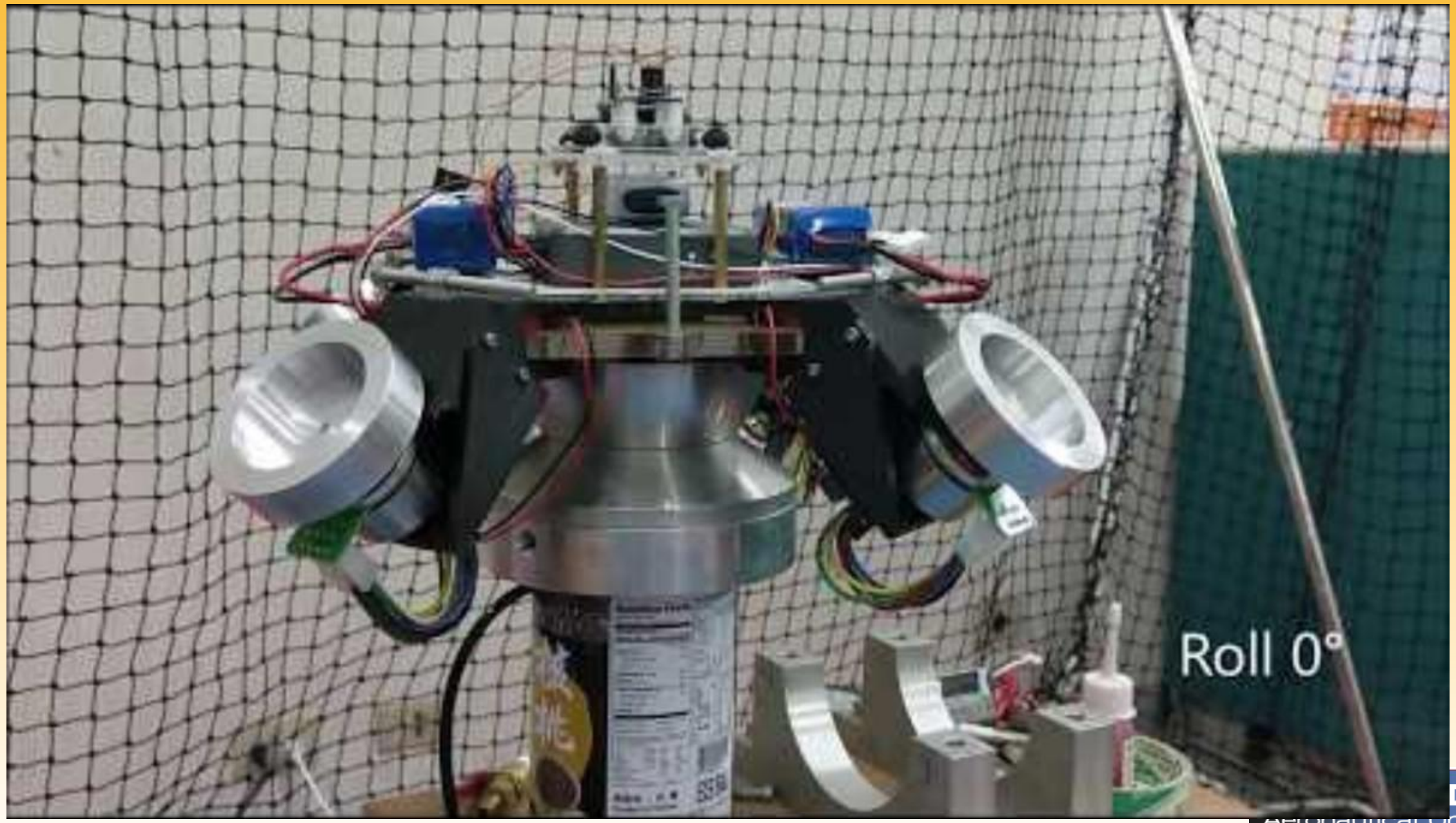


Figure 4.3.1-37. Attitude Determination and Control Subsystem (ADCS). A complete ADCS (the attitude part of an AOCS) includes a controller, actuators, the spacecraft ("the plant"), and sensors that work together to maintain or change spacecraft attitude in response to changing mission requirements.

Example - video



Questions?

- NASA, “Space Vehicle Control Systems,” online at: s3vi.ndc.nasa.gov/ssri-kb/static/resources/III.4.3.1_Space_Vehicle_Control_Systems.pdf, last accessed 3/31/2026

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